
Quotient Causal Bayesian Optimization

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Abstract

Causal Bayesian optimization (CBO) finds the intervention that optimizes a target variable, but existing methods assume the full causal DAG over all manipulable variables is known — a level of structural knowledge practitioners rarely have. We introduce Quotient CBO (QCBO), which coarsens the manipulable variables and requires only a *partition* of them into clusters together with the cluster-level graph (a C-DAG): identification, exploration-set construction, and the causal GP prior are all derived from the quotient structure. We frame this as a modeling contribution and prove the structural robustness it confers — *any* misspecification of edges inside a cluster is provably invisible to QCBO — together with a price-of-coarsening characterization: the attainable optimum degrades monotonically along the coarsening lattice, with equality exactly when an optimal intervention set is a union of clusters. On the standardized CAUSALBO benchmark (a curated set of hard-intervention SCMs, scored with the community GAP / PA-GAP metrics) and a purpose-built confounded-cluster benchmark, QCBO at the identity partition coincides with full-knowledge CBO by construction, coarse partitions trade optimality for robustness exactly as predicted, and an intra-cluster misspecification that structurally damages fine-grained CBO (deleting its optimal intervention target from the exploration set) leaves QCBO’s trajectories byte-for-byte unchanged.

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] unifies two ideas: when the variable Y we wish to optimize sits inside a causal system, the right search space is the space of *interventions* ($X_S = \mathbf{x}_S$), and observational data plus the causal graph yield informative priors over the interventional response surfaces. The catch is the input requirement. CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume the full DAG over all observables — every edge, every latent confounder — is specified correctly. Practitioners typically have something weaker and more reliable: they know the *groups*. A clinician treats the medications a patient takes (e.g. statin and aspirin) as one “treatment” lever without committing to their pharmacological interactions; a reef ecologist distinguishes nutrient/chemistry stressors from temperature/depth forcing without specifying the edges *within* either subsystem. The graph-free end of the spectrum is served by methods that learn structure online [Mukherjee et al., 2024, Branchini et al., 2023] at a heavy per-trial cost. The middle of the spectrum — cluster-level knowledge — has well-developed identification theory [Anand et al., 2023, Ferreira & Assaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method.

QCBO fills this gap. The practitioner supplies a partition π of the manipulable variables; QCBO builds the quotient graph $\mathcal{G}^\pi$ (a C-DAG over clusters, with bidirected edges absorbing latent confounding via the latent projection of Lee et al., 2019), restricts exploration to cluster-level interventions whose effects are identifiable from $\mathcal{G}^\pi$, and uses the identified cluster-level do-calculus estimates as GP prior means. The central structural insight is that this construction never looks inside a cluster:

Contributions.

- We formalize QCBO: Bayesian optimization over cluster-level interventions with identification performed on the C-DAG by a complete algorithm [Shpitser & Pearl, 2006], an identifiability-gated exploration set, and a two-tier causal GP prior (§3).
- As a *modeling* contribution (not a claim of new estimation theory), we show what the construction buys: an *intra-cluster invariance* property (Prop. 2) — every quantity QCBO computes is invariant to arbitrary edge misspecification within clusters — and a *price-of-coarsening* characterization (Prop. 5): the attainable optimum is monotone along the coarsening lattice, with an exact condition for losslessness. We also prove that validity is *not* monotone under refinement (Prop. 3: refining a valid partition can introduce a cyclic quotient), while exploration and identifiability are monotone under valid refinements (Prop. 4). The propositions have short, self-contained proofs (§4); per-dataset partitions and C-DAGs are in Appendix A.
- We evaluate on the standardized CAUSALBO benchmark [Anonymous, 2026] — a curated subset of hard-intervention SCMs (ToyGraph, Synthetic, Synthetic-2, Healthcare, Epidemiology, Ecology), scored with its GAP / PA-GAP metrics and oracle optima — and on a purpose-built confounded-cluster benchmark that separates two damage regimes of DAG misspecification: prior bias (recoverable by the BO loop) and exploration-set damage (unrecoverable). We verify that coarsening is provably immune to both when the misspecification is intra-cluster (§6).

2 Background

SCMs and interventions. A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ induces a distribution over observables $\mathbf{V}$ and, for any $S \subseteq \mathbf{V}$, interventional distributions $P(\cdot \mid (X_S = \mathbf{x}_S))$ obtained by replacing the mechanisms of X_S with constants [Pearl, 2009]. The induced graph $\mathcal{G}$ has a directed edge per functional argument and a bidirected edge per shared latent; projecting out latents yields an ADMG over $\mathbf{V}$ [Lee et al., 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(S^*, \mathbf{x}^*) = \arg \min_{\substack{S \in \mathfrak{S} \\ \mathbf{x}_S \in \mathcal{X}_S}} \mathbb{E}[Y \mid (X_S = \mathbf{x}_S)], \quad (1)$$

maintaining one GP per *candidate intervention set* S in an exploration set $\mathfrak{S}$, with prior mean given by the do-calculus estimate of $\mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]$ from observational data and the assumed graph, and selecting $(S, \mathbf{x})$ by expected improvement [Garnett, 2023]. We use “candidate intervention set” (or, loosely and only by analogy to the causal-bandit lineage that supplies the pruning below, “arm”) for a member of $\mathfrak{S}$; each is itself a *continuous* BO sub-problem over $\mathbf{x}_S$, not a single bandit arm. The exploration set is pruned structurally (MIS/POMIS; Lee & Bareinboim, 2018); both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026]). *Given a DAG $\mathcal{G}$ on $\mathbf{V}$, a **coarsening** is a partition $\pi = \{C_1, \dots, C_{d'}\}$ of $\mathbf{V}$ such that the induced quotient $\mathcal{G}^\pi$ — nodes $\{C_1, \dots, C_{d'}\}$, edge $C_k \rightarrow C_\ell$ ($k \neq \ell$) iff $\exists V_i \in C_k, V_j \in C_\ell$ with $V_i \rightarrow V_j$ in $\mathcal{G}$ — is acyclic. π is then called valid.*

For ADMGs the quotient additionally carries $C_k \leftrightarrow C_\ell$ iff some $V_i \in C_k, V_j \in C_\ell$ are connected by a bidirected edge; intra-cluster bidirected edges are dropped. The resulting object is a C-DAG in the sense of Anand et al. [2023], who prove that do-calculus on the C-DAG is sound and complete for the equivalence class of fine-grained graphs that quotient to it. Markov properties transfer to the cluster level, but faithfulness does not [Wahl et al., 2024].

3 Quotient Causal Bayesian Optimization

3.1 Setup

The practitioner partitions only the manipulable variables: π is a partition of $\mathbf{X}$, the target Y is kept as a singleton, and every non-manipulable observable remains an atomic (singleton) vertex

of the quotient. Mixing manipulable and non-manipulable variables in one cluster is disallowed — intervening on a cluster must be a physically executable action. Hidden confounders are first projected out of $\mathcal{G}$ [Lee et al., 2019], giving an ADMG $\mathcal{G}^*$ over observables; the coarsened graph $\mathcal{G}^\pi$ is the quotient of $\mathcal{G}^*$ under π (plus atomic vertices), and validity (acyclicity of the directed part) is checked. Intervening on a cluster C_k means jointly setting all of its members: ($C_k = \mathbf{c}_k$). The candidate exploration set contains all unions of at most m clusters:

$$\mathfrak{S}_0^\pi = \left\{ \bigcup_{C_k \in \mathcal{S}} C_k \mid \emptyset \neq \mathcal{S} \subseteq \pi, |\mathcal{S}| \leq m \right\}. \quad (2)$$

Remark 1 (Intervention granularity). *Merging manipulable variables coarsens the action space: QCBO can only set a cluster jointly, never a strict subset of it. This is the mechanism behind both the robustness (§4) and the price (Prop. 5) of coarsening.*

3.2 Identifiability-gated exploration and the two-tier prior

For each candidate $S \in \mathfrak{S}_0^\pi$ we pose the cluster-level query $P(Y \mid (S))$ on $\mathcal{G}^\pi$ and run the complete ID algorithm of Shpitser & Pearl [2006] (via *ananke*; Lee et al., 2023). Following the arm-pruning tradition of structural causal bandits [Lee & Bareinboim, 2018], QCBO *commits to its structural beliefs*: the exploration set $\mathfrak{S}^\pi$ keeps exactly the candidates whose effect is identifiable from $\mathcal{G}^\pi$,

$$\mathfrak{S}^\pi = \{ S \in \mathfrak{S}_0^\pi : P(Y \mid (S)) \text{ identifiable from } \mathcal{G}^\pi \}. \quad (3)$$

This gate is exact (ID is complete on the C-DAG, and identification from $\mathcal{G}^\pi$ is sound for every fine graph in its class; Anand et al., 2023), and it is what transmits structural misspecification into behaviour: a wrong assumed graph keeps or removes *arms*, not just priors (§6.3). The alternative design — exploring non-identified arms under an uninformative prior — trades this structural damage for exploration overhead; we discuss it in §8.

For $S \in \mathfrak{S}^\pi$, the GP prior mean is constructed in two tiers. The ID gate guarantees a non-parametric formula exists; our *estimator* instantiates it as a cascade of GP-based plug-ins (backdoor adjustment $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^\pi$. The cascade is sound but not complete, so on the rare queries it cannot instantiate we fall back to the observational mean:

$$\mu_0^S(\mathbf{x}_S) = \begin{cases} \hat{\mathbb{E}}[Y \mid (X_S = \mathbf{x}_S)] & \text{estimator cascade instantiates the ID formula,} \\ \bar{Y}_{\text{obs}} & \text{otherwise (logged as a diagnostic).} \end{cases} \quad (4)$$

Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data — no sub-cluster heuristic is involved.

Assumption 1 (Valid partition). *The practitioner supplies a partition π of $\mathbf{X}$ whose quotient $\mathcal{G}^\pi$ is acyclic. (Validity is checkable from cluster-level knowledge alone and is verified by the implementation.)*

Assumption 2 (Estimator-completeness boundary). *The ID gate (Eq. 3) is complete; the GP estimator cascade behind Eq. (4) is sound but not complete. Queries that pass the gate but miss the cascade are reported and receive the uninformative tier. In all benchmarks of §6 this set is empty.*

3.3 Algorithm

4 Theoretical properties

These results are a *modeling* contribution rather than new estimation theory: QCBO’s inputs are the quotient $\mathcal{G}^\pi$ and data, and the quotient is by construction blind to intra-cluster edges. The propositions below make precise what that design buys (invariance, Prop. 2), what it costs (price of coarsening, Prop. 5), and where it is subtle (validity is not refinement-monotone, Prop. 3). The proofs are short, compositional arguments about quotients and set inclusion — which is exactly why the invariance is *structural*: no intra-cluster hypothesis is even representable.

Throughout, π, π' are manipulable-only partitions as in §3.1; $\pi_\top$ denotes the identity (all-singleton) partition; π' *refines* π ($\pi' \preceq \pi$) iff every cluster of π' is contained in a cluster of π .

Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: Assumed graph $\mathcal{G}$ (or directly $\mathcal{G}^\pi$); partition π of $\mathbf{X}$; target Y ; budget T ; observational data $\mathcal{D}_{\text{obs}}$; cluster budget m .

- 1: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\pi \leftarrow$ quotient of $\mathcal{G}^*$ under π ; check acyclicity.
- 2: Build $\mathfrak{S}^\pi$ via Eqs. (2), (3) (one complete-ID call per candidate).
- 3: For each $S \in \mathfrak{S}^\pi$: fit the prior μ_0^S via Eq. (4) on $\mathcal{D}_{\text{obs}}$.
- 4: Collect initial interventional data $\mathcal{D}_0$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors (μ_t^S, k_t^S) for all $S \in \mathfrak{S}^\pi$.
- 7: $(S_t, \mathbf{x}_t) \leftarrow \arg \max_{S, \mathbf{x}} \text{EI}_{S,t}(\mathbf{x}) / \text{cost}(S, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t$.
- 9: **end for**
- 10: **return** $(S_{t^*}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

Proposition 1 (Identity recovery). *Under $\pi = \pi_\top$, $\mathcal{G}^\pi = \mathcal{G}^*$ and $\mathfrak{S}_0^\pi$ is the set of all subsets of $\mathbf{X}$ of size at most m ; QCBO reduces to CBO with an identifiability-gated exploration set.*

Proof sketch. Singleton clusters make the quotient map the identity, and unions of at most m singletons are exactly the size- $\leq m$ subsets (full proof App. B). $\square$

Proposition 2 (Invariance to intra-cluster misspecification). *Let π be valid for $\mathcal{G}$ and let $\mathcal{G}'$ differ from $\mathcal{G}$ by an arbitrary set of insertions, deletions, or reversals of edges (directed or bidirected) whose endpoints lie within a single cluster of π , with $\mathcal{G}'$ acyclic. Then $(\mathcal{G}')^\pi = \mathcal{G}^\pi$, and consequently the exploration set (Eq. 3), every identification formula, every prior (Eq. 4), and — given the same data and seeds — the entire QCBO trajectory are identical under $\mathcal{G}$ and $\mathcal{G}'$.*

Proof sketch. Latent projection touches only latent vertices, and the quotient keeps a cluster-level edge iff some fine edge crosses *distinct* clusters; so intra-cluster changes leave $\mathcal{G}^\pi$ — and every object derived from it ($\mathfrak{S}^\pi$, ID formulas, priors) — fixed (full proof App. B). $\square$

Remark 2 (What invariance buys). *Prop. 2 is a statement about the method, not about estimation error: QCBO cannot even represent an intra-cluster hypothesis, so no intra-cluster mistake can affect it. §6.3 shows the same mistakes can structurally damage fine-grained CBO by deleting its best arm.*

Proposition 3 (Validity is not refinement-monotone). *There exist a valid partition π and a refinement $\pi' \preceq \pi$ whose quotient $\mathcal{G}^{\pi'}$ is cyclic; hence refinement does not preserve validity. Validity is nonetheless decidable from cluster-level structure alone — π' is valid iff the directed part of $\mathcal{G}^{\pi'}$ is acyclic (Def. 1) — so it must be re-verified after each refinement.*

Proof sketch. The fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$ are intra-cluster under the coarsest partition but expose a 2-cycle $A \rightleftharpoons B$ once $\{a_1, a_2\}, \{b_1, b_2\}$ are split; decidability is the acyclicity clause of Def. 1 (full proof App. B). $\square$

Proposition 4 (Monotonicity under valid refinement). *Let $\pi' \preceq \pi$ with both valid, and take $m = \infty$ in Eq. (2). Then (i) $\mathfrak{S}_0^\pi \subseteq \mathfrak{S}_0^{\pi'}$; (ii) every $S \in \mathfrak{S}^\pi$ is also in $\mathfrak{S}^{\pi'}$, i.e. effects identifiable from $\mathcal{G}^\pi$ remain identifiable from $\mathcal{G}^{\pi'}$, with the same functional.*

Proof sketch. Unions of π -clusters are unions of π' -clusters, and quotienting is compositional, so a functional identifying $\mathbb{E}[Y \mid (S)]$ across the class compatible with $\mathcal{G}^\pi$ works a fortiori on the smaller class compatible with $\mathcal{G}^{\pi'}$ [Anand et al., 2023] (full proof App. B). $\square$

Proposition 5 (Price of coarsening). *Let $\mu^*(S) = \min_{\mathbf{x}_S} \mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]$ and $V(\pi) = \min_{S \in \mathfrak{S}_0^\pi} \mu^*(S)$, with $m = \infty$. For any valid $\pi' \preceq \pi$:*

$$V(\pi_\top) \leq V(\pi') \leq V(\pi),$$

and $V(\pi) = V(\pi_\top)$ iff some optimizer S^ of Eq. (1) is a union of π -clusters.*

Table 1: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior assumed	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al., 2025]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee & Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS (atomic)
QCBO (ours)	nothing	partition of $\mathbf{X}$ + C-DAG	ID-gated cluster unions

Proof sketch. The exploration sets nest, $\mathfrak{S}_0^\pi \subseteq \mathfrak{S}_0^{\pi'} \subseteq 2^{\mathbf{X}} \setminus \{\emptyset\}$, so the minima are monotone; equality holds iff the unrestricted argmin already lies in $\mathfrak{S}_0^\pi$ (full proof App. B). $\square$

Remark 3 (Damage taxonomy for misspecification). *Under the gated design (Eq. 3), a wrong assumed graph can hurt in two distinct ways. Tier-1 (prior bias): the arm survives but its prior mean is wrong; expected improvement wastes evaluations, and the interventional data eventually correct it — the damage is a convergence-speed cost. Tier-2 (arm damage): the wrong graph flips the identifiability verdict, deleting an arm from $\mathfrak{S}^\pi$ (or admitting a spurious one); if the deleted arm contained the optimum, no amount of data recovers it. Prop. 2 says coarsening is immune to both tiers whenever the misspecified edges are intra-cluster.*

5 Related work

The recent survey of Anonymous [2026] organizes the CBO landscape along four design axes — graph knowledge, intervention representation, surrogate, and decision rule — and its “unknown causal structure” family contains only methods that *learn* or *average* graphs (CEO, GACBO, Durand et al.). QCBO occupies an unfilled cell on the graph-knowledge axis: it neither assumes the full DAG nor learns one, but takes a coarser, more reliable input (a partition + C-DAG) and identifies on the quotient. It also speaks directly to two open problems the survey highlights — robustness to hidden confounding and scalable unknown-graph optimization. Concretely, methods that relax CBO’s full-graph requirement either learn structure online — GACBO maintains a posterior over plausible DAGs at substantial per-trial cost [Mukherjee et al., 2024], local-discovery CBO learns around Y [Branchini et al., 2023], and Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption and search over $\mathbf{X}$ — or assume an equivalence class is known [Park et al., 2025]. Even the most recent CBO work that broadens the action space, contextual CBO [Arsenyan et al., 2026], still assumes the full graph is given. QCBO instead assumes *less structure of a different kind*: a partition with cluster-level edges, learning no edges at all. The identification side of this regime is established — C-DAG do-calculus [Anand et al., 2023], its extension to C-DMGs over ADMGs [Ferreira & Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], and recent work on relaxing partition admissibility [Tikka et al., 2025] and on C-DAGs as background knowledge for discovery [Wang et al., 2025] — but none of these perform optimization.

Multi-fidelity and abstraction-based BO. Closest in spirit is the concurrent position paper of Zeitler [2025], which proposes MFACBO — a hybrid of CBO and multi-information-source BO [Poloczek et al., 2017] unified through approximate causal abstraction [Beckers & Halpern, 2019]. MFACBO coarsens the *outcome* (Y_{low} as a cheap, biased proxy of Y_{high}); QCBO is complementary in that it coarsens the *manipulable inputs* into cluster-level interventions and exploits identification on the resulting C-DAG. The two axes are orthogonal and could be combined in a single surrogate; we leave the hybrid to future work.

6 Experiments

6.1 Setup

Standardized benchmark. Our main evaluation uses the curated hard-intervention subset of the CAUSALBO benchmark [Anonymous, 2026]: *ToyGraph* and *Synthetic-2* (chains $X \rightarrow Z \rightarrow Y$),

Table 2: QCBO on the curated CAUSALBO benchmark: final Y , GAP, and PA-GAP (mean $\pm$ s.e., 5 seeds, 100 trials), scored by the benchmark’s own pipeline against its oracle y^* . QCBO-finest is CBO on the full DAG (Prop. 1); QCBO-coarse uses the domain partition of Appendix A. GAP is improvement relative to each method’s initial incumbent and y^* is an extreme oracle reference, so the values read as a relative ranking rather than an attainment fraction (see Limitations). Auto-generated; the included file ends with its own `\bottomrule`.

Dataset	Method	Final Y	GAP	PA-GAP
ToyGraph	CBO (=QCBO-finest)	-2.172 ± 0.000	0.390 ± 0.056	0.176 ± 0.065
	QCBO-coarse	-2.172 ± 0.000	0.453 ± 0.068	0.238 ± 0.058
Synthetic-2	CBO (=QCBO-finest)	-1.856 ± 0.000	0.370 ± 0.036	0.082 ± 0.029
	QCBO-coarse	-0.511 ± 0.000	0.490 ± 0.007	0.018 ± 0.006
Synthetic	CBO (=QCBO-finest)	-1.739 ± 0.212	0.127 ± 0.109	0.018 ± 0.017
	QCBO-coarse	-1.737 ± 0.257	0.317 ± 0.099	0.051 ± 0.015
Healthcare	CBO (=QCBO-finest)	5.352 ± 0.041	0.208 ± 0.128	0.011 ± 0.009
	QCBO-coarse	5.329 ± 0.015	0.360 ± 0.071	0.033 ± 0.010
Epidemiology	CBO (=QCBO-finest)	-2.344 ± 0.031	0.542 ± 0.106	0.168 ± 0.044
	QCBO-coarse	-2.300 ± 0.026	0.473 ± 0.058	0.254 ± 0.027
Ecology	CBO (=QCBO-finest)	5.023 ± 0.036	0.099 ± 0.067	0.001 ± 0.001
	QCBO-coarse	4.983 ± 0.051	0.286 ± 0.083	0.020 ± 0.008

Synthetic (the six-observable graph of Aglietti et al., 2020 with latents $U_1 \rightarrow \{A, Y\}$, $U_2 \rightarrow \{B, Y\}$; manipulable $\{B, D, E\}$), *Healthcare* (statin/aspirin $\rightarrow$ PSA, with observed age/BMI confounding), *Epidemiology* (an HIV SCM, manipulable $\{L, B\}$), and *Ecology* (a fitted coral-reef SCM, manipulable $\{N, O, C, T, D\}$). We run each through the benchmark’s own CBO loop and score the exported best-so-far trajectories with its *GAP* and *PA-GAP* metrics against its shipped oracle optima y^* — so our numbers are directly comparable to the benchmark’s other methods. For each dataset we use the identity partition (QCBO-finest, which by Prop. 1 is CBO) and one domain-meaningful coarse partition; the partitions and C-DAGs are listed in Appendix A.

Controlled robustness benchmarks. The standardized datasets assume causal sufficiency or place confounders away from manipulable pairs, so none exhibits the structure that drives misspecification damage. We therefore use two controlled SCMs for the robustness study: *CompleteGraph* (=Synthetic, for Tier-1 prior bias) and a purpose-built *ConfoundedCluster* “bow” benchmark (for Tier-2 arm deletion): manipulable $\{A, B, C\}$, latent $U \rightarrow \{B, C\}$ giving $\text{corr}(B, C) \approx 0.70$, with the strongest lever (B).

Protocol. On the standardized benchmark we follow its protocol ($T=100$ trials, unit costs, the single-draw SEM objective and offline y^*), report GAP / PA-GAP at 100/50/20 trials over 5 seeds, and route every method’s trajectory through the same scorer. The exploration-set cap matches the benchmark’s own ($m = \min(5, |\mathbf{X}|)$): all interventions on up to m manipulable variables), so at $\pi_\top$ QCBO-finest searches exactly CBO’s candidate set. On the controlled benchmarks all methods share the same seeds, initial-data budget, trial budget, and per-benchmark cluster cap m ; a fairness check enforces identical trajectory lengths, seed coverage, and initial-budget across methods. Reported uncertainties are standard errors (sample s.d., $\text{ddof}=1$).

6.2 Main comparison and the price of coarsening

Table 2 reports QCBO on the curated CAUSALBO benchmark under its own GAP / PA-GAP scoring. Two things to read off it. First, at the identity partition QCBO is the CBO algorithm on the full DAG (Prop. 1): the quotient is the original graph, so the exploration set is CBO’s size- $\leq m$ candidate family and each arm uses the identifiability-gated estimator of §3.2. We therefore report QCBO-finest as the single full-knowledge anchor — an algorithmic equivalence at $\pi_\top$, not a numerical match to any separately implemented CBO baseline — and locate coarsening’s value on the price and robustness axes rather than in a spurious convergence “win” at $\pi_\top$. Second, the finest $\rightarrow$ coarse gap on each dataset is exactly the *price of coarsening* (Prop. 5): it is small or zero

when an optimal intervention is (a union of) clusters — e.g. the deterministic chains, where the cluster intervention recovers the optimum — and grows when a coarse cluster must be set jointly even though the optimum splits it.

The controlled CompleteGraph case makes the price exact. By brute-force Monte Carlo on its SEM, the unique optimum is the *joint* arm $\{B, D\}$ with $\mu^*(\{B, D\}) = -3.60$, while additionally forcing E destroys it ($\mu^*(\{B, D, E\}) = -0.57$) — a textbook “where to intervene” instance [Lee & Bareinboim, 2018]: left free, $E = f(A, \cdot)$ inherits U_1 through A and partially compensates U_1 ’s direct effect on Y . The coarse partition $\{B\}, \{D, E\}$ therefore sits at its Prop. 5 floor $V(\pi) = \mu^*(\{B\}) = -1.31$: the optimal action (B, D) requires splitting the $\{D, E\}$ cluster and is simply not representable. The price is merge-specific, not a function of fineness alone — Prop. 5 gives the exact losslessness condition.

Coarsening can also accelerate search. A third effect is visible in Table 2’s GAP column. The identity partition carries CBO’s full size- $\leq m$ candidate family, so on the higher-dimensional datasets it spreads a fixed 100-trial budget across many arms (seven for *Synthetic*, thirty-one for *Ecology*), which slows convergence; the coarse partition keeps far fewer cluster-level arms and concentrates the budget. Consequently QCBO-coarse attains a *higher* GAP than QCBO-fine ($\equiv$ CBO) on *ToyGraph*, *Synthetic*, *Healthcare*, and *Ecology* while reaching comparable final Y . Coarsening thus offers a sample-efficiency benefit that is orthogonal to — and, on these benchmarks, more often realized than — the price it can charge in attainable optimum (Prop. 5): the trade is favourable whenever the optimum does not require splitting a cluster.

6.3 Robustness: two damage regimes

We separate the two tiers of Remark 3; Figure 1 shows the corresponding trajectories.

Tier-1 (prior bias) — CompleteGraph (Table 3). Deleting the inter-cluster edge $B \rightarrow C$ (NoBC) or the edge $C \rightarrow D$ (NoCD) leaves every identifiability verdict intact — no arm is added or removed, so $|\mathfrak{S}^\pi|$ is unchanged — but corrupts the priors of the affected arms. This is a *convergence-speed* cost, not a structural one: expected improvement wastes early evaluations on the mis-primed arms, but the interventional data progressively correct the prior. By the 100-trial budget the fine partition’s paired gap to the correctly-specified run is statistically indistinguishable from zero under both conditions ($+0.11 \pm 0.11$ for NoBC, -0.05 ± 0.06 for NoCD; Table 3) — the prior bias is absorbed rather than locked in, in sharp contrast to the Tier-2 arm deletion below, which never recovers. The coarse partition $\{B\}, \{D, E\}$ is immune to both: each misspecified edge either lies inside a cluster or is made redundant at the quotient level ($C \rightarrow E$ keeps the $\{C\} \rightarrow \{D, E\}$ cluster edge alive when $C \rightarrow D$ is deleted), so by Prop. 2 its trajectory is unchanged and it sits at its Prop. 5 floor.

Tier-2 (arm damage) — ConfoundedCluster (Table 4). The misspecified DAG WrongBC adds a single spurious intra-cluster edge $B \rightarrow C$. Together with the (real) latent confounding $B \leftrightarrow C$ this forms a bow, so $P(Y \mid (B))$ is not identifiable from the assumed fine graph: the identity-partition exploration set provably loses its best target $\{B\}$ (ground-truth arm values: $\mu^*(\{B\}) = -3.75$, vs. $\mu^*(\{C\}) = -2.75$ for the best *surviving* singleton). No amount of interventional data restores a deleted target, and the experiment shows exactly that: the fine-grained method’s paired gap is $\approx +1.0$ at every seed. Under the coarse partition $\{A\}, \{B, C\}$ the bow is intra-cluster: Prop. 2 applies, and the QCBO trajectories under the correct and misspecified DAGs are *byte-for-byte identical* at every seed (the cluster target (B, C) also attains $\mu^*(\{B, C\}) = -4.25$, dominating every singleton).

7 Discussion

- Prop. 2 is structural, not statistical: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 5) for immunity to an entire class of misspecifications. The practitioner chooses where on this dial to sit by choosing how much structure to commit to.
- The damage taxonomy (Remark 3) reframes “is CBO robust to a wrong graph?”: prior bias is self-healing under any reasonable acquisition; arm damage is not. Robustness analyses that only perturb priors miss the failure mode that matters.
- The partition need not be guessed once and forever: Prop. 4 makes refinement monotonically safe (identifiable arms are never lost), although validity must be re-checked (Prop. 3). Discovering

Table 3: Tier-1 robustness (CompleteGraph; 5 seeds): prior-bias damage is partially self-healing for the fine partition and absent for the coarse one.

Partition	Assumed DAG	$ \mathcal{G}^\pi $	Final Y	Gap vs. correct
finest	NoBC	7	-3.404 ± 0.167	$+0.112 \pm 0.113$
finest	NoCD	7	-3.571 ± 0.002	-0.054 ± 0.055
finest	correct	7	-3.516 ± 0.054	—
$\{B\}, \{DE\}$	NoBC	3	-1.179 ± 0.137	$+0.000 \pm 0.000$
$\{B\}, \{DE\}$	NoCD	3	-1.179 ± 0.137	$+0.000 \pm 0.000$
$\{B\}, \{DE\}$	correct	3	-1.179 ± 0.137	—

Table 4: Tier-2 robustness (ConfoundedCluster bow benchmark; 5 seeds). “Gap” is the paired per-seed difference to the same partition under the correct DAG. WrongBC deletes the optimal target $\{B\}$ at the finest partition (a $\approx +1.0$ unrecoverable gap); ± 0.000 for the coarse partition reflects byte-identical trajectories (Prop. 2).

Partition	Assumed DAG	$ \mathcal{G}^\pi $	Final Y	Gap vs. correct
finest	WrongBC	2	-2.749 ± 0.000	$+1.002 \pm 0.000$
finest	correct	3	-3.750 ± 0.000	—
$\{A\}, \{BC\}$	WrongBC	2	-4.232 ± 0.006	$+0.000 \pm 0.000$
$\{A\}, \{BC\}$	correct	2	-4.232 ± 0.006	—

the partition online from interventional data — e.g. with the RePaRe routine of Madaleno et al. [2026] — is a natural recursive extension that we develop in follow-up work.

8 Limitations

(i) The gated design (Eq. 3) commits to the assumed structure; the un-gated variant (explore non-identified arms under an uninformative prior) converts arm damage into exploration overhead and deserves a systematic comparison. (ii) The GP estimator cascade behind Eq. (4) is sound but not complete (Assumption 2). (iii) We evaluate on the curated hard-intervention subset of the CAUSALBO benchmark plus a purpose-built confounded-cluster benchmark; soft-intervention / function-network settings are out of scope (QCBO is a hard-intervention method). (iv) Our latent projection handles the bidirected structure these benchmarks exhibit; bidirected-input projection through drop-only chains is not implemented (it does not arise here), so the “absorbs latent confounding” claim is scoped to that setting. (v) Faithfulness does not transfer to the cluster level [Wahl et al., 2024], so cluster-level conditional independences observed in data must not be read back as fine-grained structure. (vi) We evaluate at unit intervention costs; cost-asymmetric settings change where coarsening pays. (vii) The benchmark’s oracle optima y^* are extreme reference values — in some datasets attained by interventions outside the domain the methods search (e.g. Epidemiology, whose reported optimizer lies far outside the interventional ranges) — so the absolute GAP is uniformly deflated across all methods; we therefore read GAP and PA-GAP as relative rankings, not attainment fractions, and locate QCBO’s contribution in the robustness and price analyses rather than in benchmark GAP magnitudes.

9 Conclusion

QCBO makes causal Bayesian optimization usable at the level of structural knowledge practitioners actually have — clusters — with a complete identification gate, a provable invariance to intra-cluster misspecification, and an exact characterization of what coarseness costs. On the spectrum from structure-free BO to oracle-graph CBO, it provides the missing, principled middle; refining the partition as evidence accumulates closes the remaining gap and is the subject of ongoing work.

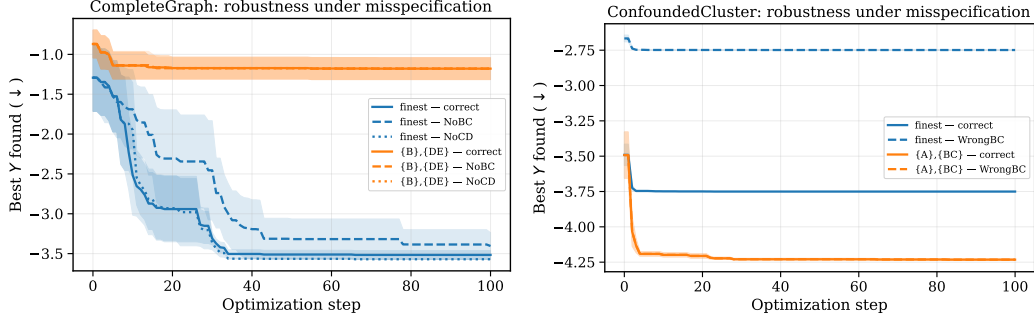


Figure 1: Robustness under misspecification (best-so-far Y , mean $\pm$ s.e.): Tier-1 prior bias on CompleteGraph (left) and Tier-2 arm deletion on the ConfoundedCluster bow benchmark (right). Colour = partition, solid = correct DAG, dashed = misspecified DAG. At the coarse partition the misspecified and correct curves coincide (Prop. 2); at the finest partition the bow benchmark’s WrongBC curve is strictly worse, reflecting the deleted optimal target.

A Benchmark datasets and partitions

For each curated CAUSALBO dataset we run QCBO at the identity partition (QCBO-finest $\equiv$ CBO, Prop. 1) and at one domain-meaningful coarse partition. The coarse partition groups manipulable variables a practitioner would lump into a subsystem without committing to the edges *within* it; validity (acyclicity of the quotient) is checked automatically (Assumption 1). Table 5 lists the manipulable set and the coarse partition per dataset; the DAGs are read from each dataset’s structural equations. Figures 2–7 draw each dataset’s fine-grained ADMG beside its quotient C-DAG. Throughout, solid arrows denote directed causes and dashed red double arrows denote latent confounders (bidirected edges); the target Y is gold, manipulable variables orange, other observables blue, and a rounded box marks a quotient cluster of more than one variable.

Table 5: Curated benchmark datasets: manipulable variables and the coarse partition used (the identity partition is always also run). Y is the target; non-manipulable observables remain atomic singletons of the quotient.

Dataset	Manipulable	Coarse partition	Rationale
ToyGraph / Synthetic-2	$\{X, Z\}$	$\{X, Z\}$	absorb the chain $X \rightarrow Z$
Synthetic (CompleteGraph)	$\{B, D, E\}$	$\{B\}, \{D, E\}$	D, E are the output stage
Healthcare	$\{\text{Aspirin}, \text{Statin}\}$	$\{\text{Aspirin}, \text{Statin}\}$	one “medication” lever
Epidemiology	$\{L, B\}$	$\{L, B\}$	joint exposure
Ecology	$\{N, O, C, T, D\}$	$\{C, N, O\}, \{D, T\}$	nutrient/chemistry vs. temperature/depth
ConfoundedCluster (bow)	$\{A, B, C\}$	$\{A\}, \{B, C\}$	confounded pair B, C

B Proofs

The main text (§4) states each result with a short sketch; here are the full proofs. Notation is as in §3.1 and §4.

Proof of Prop. 1. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, and unions of at most m singletons are exactly the subsets of size at most m . $\square$

Proof of Prop. 2. The latent projection removes only latent vertices; an edge between two observables is never an interior vertex of a projection path, so $(\mathcal{G}')^*$ and $\mathcal{G}^*$ differ exactly by the modified intra-cluster edges. The quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two *distinct* clusters; edges within a cluster contribute nothing. Hence the quotients coincide. All remaining objects — $\mathfrak{S}^\pi$, ID formulas, priors — are functions of $\mathcal{G}^\pi$ and data alone. $\square$

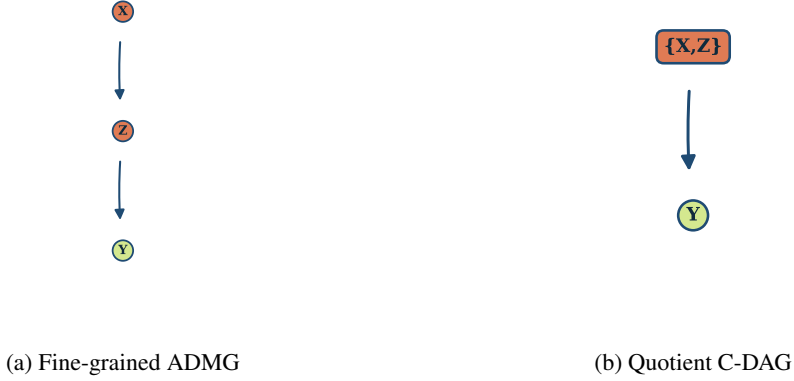


Figure 2: **ToyGraph / Synthetic-2**. The chain $X \rightarrow Z \rightarrow Y$. Merging the manipulable variables into one cluster $\{X, Z\}$ leaves a single quotient edge $\{X, Z\} \rightarrow Y$.

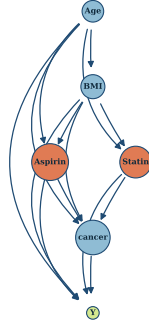


Figure 3: **Synthetic (CompleteGraph)**. Latent confounders $U_1 \rightarrow \{A, Y\}$ and $U_2 \rightarrow \{B, Y\}$ appear as the bidirected edges $A \leftrightarrow Y$ and $B \leftrightarrow Y$. Coarsening the output stage $\{D, E\}$ does not touch these confounders—they cross cluster boundaries—so both survive in the C-DAG.

Proof of Prop. 3. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement $\pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster — $a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$ — inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\pi'}$ is cyclic and π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Prop. 4. (i) A union of π -clusters is a union of π' -clusters because each π -cluster is. (ii) Quotienting is compositional: $\mathcal{G}^\pi$ is the quotient of $\mathcal{G}^{\pi'}$ under the induced partition of π' -clusters, so every fine graph compatible with $\mathcal{G}^{\pi'}$ is compatible with $\mathcal{G}^\pi$. Identifiability from $\mathcal{G}^\pi$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid (S)]$ across all SCMs whose graph is compatible with $\mathcal{G}^\pi$ [Anand et al., 2023]; a fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\pi'}$. $\square$

Proof of Prop. 5. By Prop. 4(i), $\mathfrak{S}_0^\pi \subseteq \mathfrak{S}_0^{\pi'} \subseteq \mathfrak{S}_0^{\pi^\top} = 2^{\mathbf{X}} \setminus \{\emptyset\}$; minima over larger sets are no larger. Equality holds iff the unrestricted argmin intersects $\mathfrak{S}_0^\pi$, which is exactly the stated condition. $\square$

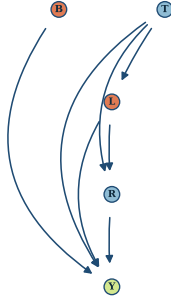


(a) Fine-grained ADMG

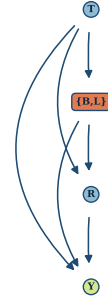


(b) Quotient C-DAG

Figure 4: **Healthcare.** Age and BMI confound the two medication levers, the cancer mediator, and the outcome (no latent confounders). The coarse partition merges Aspirin and Statin into a single “medication” cluster.



(a) Fine-grained ADMG



(b) Quotient C-DAG

Figure 5: **Epidemiology.** Treatment T drives the exposures $\{L, B\}$, the mediator R , and the outcome. The coarse partition treats the joint exposure $\{L, B\}$ as one lever.

References

- T. V. Anand, A. H. Ribeiro, J. Tian, and E. Bareinboim. Causal effect identification in cluster DAGs. In *AAAI*, volume 37, pp. 12172–12179, 2023.
- V. Aglietti, X. Lu, A. Paleyes, and J. González. Causal Bayesian optimization. In *AISTATS*, volume 108 of *PMLR*, pp. 3155–3164, 2020.
- V. Aglietti, N. Dhir, J. González, and T. Damoulas. Dynamic causal Bayesian optimization. In *NeurIPS*, volume 34, 2021.
- V. Aglietti, A. Malek, I. Ktena, and S. Chiappa. Constrained causal Bayesian optimization. In *ICML*, 2023.
- S. Beckers and J. Y. Halpern. Abstracting causal models. In *AAAI*, 2019.
- N. Branchini, V. Aglietti, N. Dhir, and T. Damoulas. Causal Bayesian optimization via local causal discovery. In *CLeaR*, volume 236 of *PMLR*, 2023.
- E. Ferreira and C. K. Assaad. Identifying macro causal effects in a C-DMG over ADMGs. *arXiv:2504.01551*, 2025.

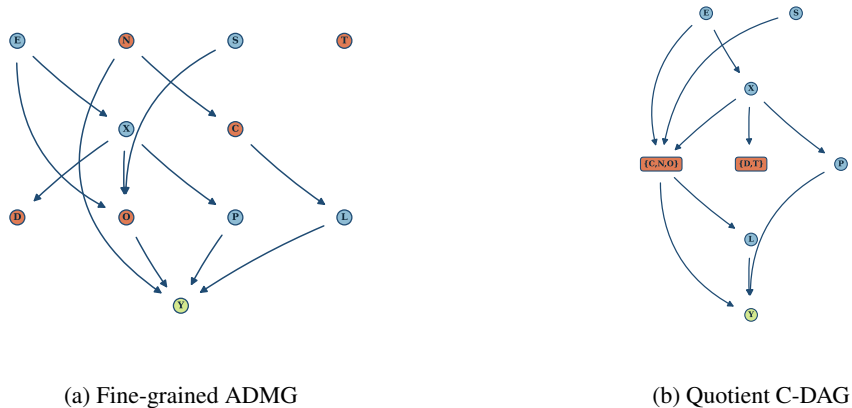


Figure 6: **Ecology**. The largest curated graph. The quotient cluster $\{D, T\}$ has *no* directed path to Y (D is a pure sink and T is isolated): the practitioner-chosen grouping cleanly isolates a subsystem irrelevant to the objective.

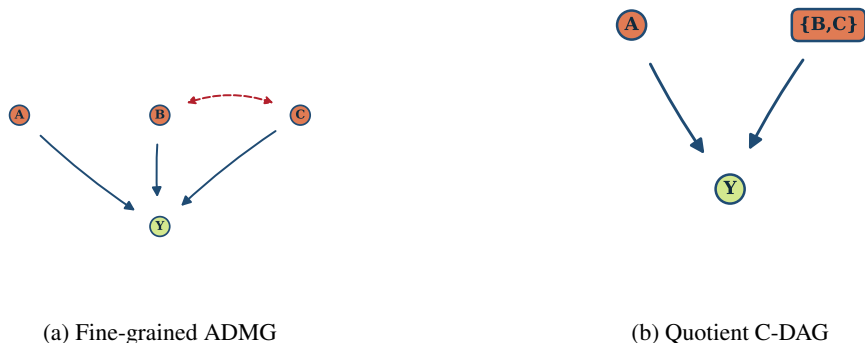


Figure 7: **ConfoundedCluster (bow)**. The latent confounder over the pair $\{B, C\}$ is a *bow*: the bidirected edge $B \leftrightarrow C$. Once $\{B, C\}$ is a single cluster the bow becomes intra-cluster and is projected out—the C-DAG carries *no* bidirected edge, the structural mechanism behind the within-cluster invariance of Prop. 2.

J. L. Gamella, J. Peters, and P. Bühlmann. Causal chambers as a real-world physical testbed for AI methodology. *Nature Machine Intelligence*, 7:107–118, 2025.

R. Garnett. *Bayesian Optimization*. Cambridge University Press, 2023.

J. J. R. Lee, R. Bhattacharya, R. Nabi, and I. Shpitser. Ananke: A python package for causal inference using graphical models. *arXiv:2301.11477*, 2023.

S. Lee and E. Bareinboim. Structural causal bandits: Where to intervene? In *NeurIPS*, 2018.

S. Lee, J. Correa, and E. Bareinboim. General identifiability with arbitrary surrogate experiments. In *UAI*, 2019.

F. Madaleno, P. Misra, and A. Markham. Coarsening causal DAG models. *arXiv:2601.10531*, 2026.

S. Mukherjee, et al. Graph-agnostic causal Bayesian optimization. *arXiv*, 2024.

J. Park, M. Zemlenyi, and V. Aglietti. Causal Bayesian optimization under partial observability. *arXiv*, 2025.

- J. Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2nd edition, 2009.
- M. Poloczek, J. Wang, and P. Frazier. Multi-information source optimization. In *NeurIPS*, 2017.
- I. Shpitser and J. Pearl. Identification of joint interventional distributions in recursive semi-Markovian causal models. In *AAAI*, 2006.
- S. Sussex, A. Makarova, and A. Krause. Model-based causal Bayesian optimization. In *ICLR*, 2023.
- Anonymous. Relaxing partition admissibility in cluster-DAGs: a causal calculus with arbitrary variable clustering. *arXiv:2511.01396*, 2025.
- J. Wahl, U. Ninad, and J. Runge. Foundations of causal discovery on groups of variables. *Journal of Causal Inference*, 12(1), 2024.
- Anonymous. Cluster-DAGs as powerful background knowledge for causal discovery. *arXiv:2512.10032*, 2025.
- J. Zeitler. Towards MFACBO: Multi-fidelity abstraction causal Bayesian optimization. In *1st Workshop on Causal Abstractions and Representations (CAR) at UAI*, 2025.
- J. Tian and J. Pearl. A general identification condition for causal effects. In *AAAI*, 2002.
- Anonymous. Causal Bayesian optimization: Foundations, methods, and applications. Under review, 2026.
- J. Durand, Y. Annadani, S. Bauer, and S. Parbhoo. Causal Bayesian optimization with unknown causal graphs. In *ICLR*, 2025.
- V. Arsenyan, A. Grosnit, H. Bou Ammar, and A. S. Dalalyan. Contextual causal Bayesian optimization. In *ICLR*, 2026.